$$\underset{0.4-\alpha}{CH_3COOH(aq)} \rightleftharpoons \underset{\alpha}{CH_3COO^-(aq)} + \underset{\alpha}{H^+(aq)}$$

$$\underset{0.4}{CH_3COONa} \rightarrow \underset{0.4}{CH_3COO^-} + \underset{0.4}{Na^+}$$

$$\underset{(0.02)}{CH_3COO^-} + \underset{0.02}{HCl} \rightarrow \underset{0.02}{CH_3COOH} + \underset{0.02}{Cl^-}$$

$\therefore$ $[CH_3COOH] = 0.4 - \alpha + 0.02 = 0.42 - \alpha \simeq 0.42$

$[CH_3COO^-] = 0.4 + \alpha - 0.02 = 0.38 + \alpha \simeq 0.38$

$$[H^+] = \frac{(1.8 \times 10^{-5})(0.42)}{(0.38)}$$

$[H^+] = 1.99 \times 10^{-5}$

$pH = - \log (1.99 \times 10^{-5})$

$= 5 - \log 1.99$

$= 5 - 0.30$

$= 4.70$

Evaluate yourself – 9

a) answer

$$pOH = pK_b + \log \frac{[salt]}{[base]}$$

We know that

$pH + pOH = 14$

$\therefore 9 + pOH = 14$

$\Rightarrow pOH = 14 - 9 = 5$

$$5 = 4.7 + \log \frac{[NH_4Cl]}{[NH_4OH]}$$

$$0.3 = \log \frac{[NH_4Cl]}{0.1}$$

$$\frac{[NH_4Cl]}{0.1} = \text{antilog of } (0.3)$$

$[NH_4Cl] = 0.1M \times 1.995$

$= 0.1995\ M$

$= 0.2M$

Amount of NH_4Cl required to prepare 1 litre 0.2M solution = Strength of NH_4Cl × molar mass of NH_4Cl

$= 0.2 \times 53.5$

$= 10.70\ g$

10.70 g ammonium chloride is dissolved in water and the solution is made up to one litre to get 0.2M solution. On mixing equal volume of the given NH_4OH solution and the prepared NH_4Cl solution will give a buffer solution with required pH value (pH = 9).

b)answer

$$pH = pK_a + \log \frac{[salt]}{[acid]}$$

$$4 = 3.75 + \log \frac{[\text{sodium formate}]}{[\text{formic acid}]}$$

[Sodium formate] = number of moles of HCOONa

$= 0.6 \times V \times 10^{-3}$

[formic acid] = number of moles of HCOOH

$= 0.8 \times 100 \times 10^{-3}$

$= 80 \times 10^{-3}$

$$4 = 3.75 + \log \frac{0.6V}{80}$$

$$0.25 = \log \frac{0.6V}{80}$$

$$\text{antilog of } 0.25 = \frac{0.6V}{80}$$

$0.6V = 1.778 \times 80$

$= 1.78 \times 80$

$= 142.4$

$$V = \frac{142.4\ mL}{0.6} = 237.33mL$$

Evaluate yourself - 10

Sodium carbonate is a salt of weak acid, H_2CO_3 and a strong base, NaOH, and hence the solution is alkaline due to hydrolysis.

$$Na_2CO_3(aq) \rightarrow 2Na^+(aq) + CO_3^{2-}(aq)$$

$$CO_3^{2-}(aq) + H_2O(l) \rightleftharpoons HCO_3^- + OH^-$$

i) $$h = \sqrt{\frac{K_w}{K_a \times C}}$$

$$= \sqrt{\frac{1 \times 10^{-14}}{5.5 \times 10^{-11} \times 0.05}}$$

$h = 6.03 \times 10^{-2}$

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